COMSATS University Islamabad Lahore Campus

Mid-Term Examination – SPRING 2023

Course Title:	Database Systems I			Course Code:	CSC371	Credit	3+1
Course	Abdul Qayyum			Programme	BSSE		
Semester:	4th	Batch:	FA21	Section:	A,B	Date:	03-05-2023
Time Allowed:	90 Minutes			Maximum Marks:			50
Student's				Reg. No.			

Important Instructions / Guidelines:

Question 1: [CLO-1] [8]

What are the limitations of File Based Approach. Explain each with an example.

Question 2: [CLO-2] [6]

Explain the External level and Conceptual level of The Three-Level ANSI-SPARC Architecture.

Question 3: [CLO-2] [6]

Draw and *explain* the job done at each tier in Three-Tier Client–Server Architecture.

Question 4: [CLO-2] [5]

Define the following terms:

a. Primary key
b. Candidate key
c. Relational database
d. Cardinality
e. Referential Integrity

Question 5: [CLO-3] [25]

Consider the following case to draw an ERD:

A Country bus Company owns several buses. Each bus is allocated to a particular route, although some routes may have several buses. Each route passes through several towns. Some of the towns have a garage where buses are kept and each of the buses are identified by the registration number and can carry different numbers of passengers, since the vehicles vary in size and can be single or double-decked. Each route is identified by a route number and information is available on the average number of passengers carried per day for each route. Drivers have an employee number, name, address, and sometimes a telephone number.

Construct/Draw an ER diagram to represent the data requirements as following:

- Identify the main entity types. [07]
- Identify the main relationship types between the entity types. [05]
- Identify attributes and associate them with entity or relationship types. [05]
- Determine candidate and primary key attributes for each entity type. [03]
- Identify the occurrence (1 - 1, 1 - *, * - *) between entities. [05]
- State any assumptions necessary to support your design.

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